

6 Probability

What students need to learn:

Ref	Content descriptor	Concepts and skills
SP m	Understand and use the vocabulary of probability and probability scale	• Distinguish between events which are; impossible, unlikely, even chance, likely, and certain to occur • Mark events and/or probabilities on a probability scale of 0 to 1 • Write probabilities in words, fractions, decimals and percentages
SP n	Understand and use estimates or measures of probability from theoretical models (including equally likely outcomes), or from relative frequency	• Find the probability of an event happening using theoretical probability • Find the probability of an event happening using relative frequency • Estimate the number of times an event will occur, given the probability and the number of trials • Use theoretical models to include outcomes using dice, spinners, coins
SP o	List all outcomes for single events, and for two successive events, in a systematic way and derive relative probabilities	• List all outcomes for single events systematically • List all outcomes for two successive events systematically • Use and draw sample space diagrams
SP p	Identify different mutually exclusive outcomes and know that the sum of the probabilities of all these outcomes is 1	• Add simple probabilities • Identify different mutually exclusive outcomes and know that the sum of the probabilities of all these outcomes is 1 • Use $1 - p$ as the probability of an event not occurring where p is the probability of the event occurring • Find a missing probability from a list or table
SP s	Compare experimental data and theoretical probabilities	• Compare experimental data and theoretical probabilities
SP t	Understand that if they repeat an experiment, they may – and usually will – get different outcomes, and that increasing sample size generally leads to better estimates of probability and population characteristics	• Compare relative frequencies from samples of different sizes

Higher

Overview

Content overview

This qualification contains:

- 1. Number**
- 2. Algebra**
- 3. Geometry**
- 4. Measure**
- 5. Statistics**
- 6. Probability**

Assessment overview

- Each paper contributes 50% of the qualification
- Two written papers
- Each paper lasts 1 hour 45 minutes
- Each paper contains 100 marks
- Paper 1 Non-calculator, Paper 2 Calculator
- Grades A*-D available (E allowed)
- Available June and November
- 20-30% of each paper assesses the functional elements of Mathematics

References

Each topic in this qualification contains a specification reference (for example, **SP a** for Statement a, Statistics and Probability), the content descriptor and examples of concepts and skills associated with that content descriptor.

Content that is Higher Tier only is indicated in **Bold type**.

The content of Foundation Tier is subsumed in the content for Higher Tier.